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BUS 303

Dr. Foy

Signature Term Project

Reflection

Before taking this course, I expected Python to be more focused on memorizing code. I work in Excel regularly, to organize information and review data, and have limited experience coding in a business analysis context. Since this was a 7-week course, we covered a large amount of information in a short time. I learned that programming is not necessarily about memorizing commands, but understanding the problem, and breaking it into smaller, attainable steps.

I believe the course and Signature Term Project achieved the main learning objectives by giving me practical experience importing, examining, cleaning, validating, and exporting business data. The Starbucks and Dunkin' datasets looked organized at first, but each contained problems that could have affected the analysis. These problems included blank rows, missing values, incorrect data types, duplicate records, irrelevant food categories, misspelled column names, and special characters that hid missing information. Working through these issues showed me that data should never be assumed to be accurate or ready for analysis simply because it opens correctly in a spreadsheet or dataframe.

One of my takeaways is the importance of validating each cleaning decision instead of making several changes and assuming they worked. For example, in the Starbucks dataset, the word "zero" caused the Sugar column to be stored as text instead of numerical data. I had to replace that value, convert the column to a float, and then confirm that every value could be treated as a number. In the Dunkin' Donuts dataset, the first missing-value check reported zero missing values. Further inspection showed that some missing information was represented by nonbreaking space characters instead of standard NaN values. After removing the special characters and converting the empty strings to NaN, three incomplete records became visible. That was a good example of why an initial output does not always tell the whole story.

The problem-solving process was one of the most valuable parts of the project. I learned to create copies of dataframes before making major changes, inspect unique values, check data types, count missing values and duplicates, and verify the shape of the dataset after removing records. When something did not work, I had to look at the error message, check spelling and punctuation, and determine whether I had run the notebook cells in the correct order. Jupyter Notebook was helpful because it allowed me to see the code, explanation, and output together. At the same time, the order of the cells could be challenging because later code depended on earlier dataframes being created correctly.

Some parts of Python became easier once I understood the basic patterns. Commands such as head(), info(), isnull().sum(), dropna(), duplicated(), and to_csv() became easier to recognize and apply. Filtering records and creating lists of categories also became more understandable with practice. The more difficult parts were keeping track of multiple dataframe names, understanding why apparently numerical columns were classified as objects, and identifying invisible special characters. Syntax was also challenging because a missing quotation mark, parenthesis, bracket, or incorrect column name could prevent the code from running.

DataCamp helped introduce the commands and allowed me to practice them in smaller exercises. It was useful because the exercises provided immediate feedback and focused on one concept at a time. However, the Signature Term Project required more independent thinking. In DataCamp, the dataset and much of the structure were already provided. In the project, I had to follow a longer process, understand why each step was necessary, and make sure that changes made in one task did not create problems later. The combination of guided DataCamp exercises and the larger Jupyter project helped reinforce the material.

My perception of Python has changed because I now see how it can support business work instead of viewing it only as a programming language. In my work in construction estimating, I regularly use Excel and work with large amounts of information from drawings, estimates, bid forms, schedules, and vendor data. This information may come from different sources and may not always be organized consistently. Python could potentially help clean repeated datasets, standardize category names, identify missing information, compare lists, and automate tasks that would otherwise require manual Excel work. It could also create a repeatable record of the steps used to process information, which is valuable when someone else needs to understand or verify the workflow.

The hints were very important, especially because this was my first time completing a full project in Anaconda and Jupyter Notebook. They made the project less challenging, but I do not think that reduced the learning value. The hints helped identify the correct direction while I still had to enter the code, run it, interpret the results, and correct errors. Without the hints, I probably could have completed some parts through additional research, but it would have been difficult to complete the full project confidently within a seven-week course. As I gained more experience, I became better able to understand what the hints were asking me to accomplish instead of simply following them.

Before this course, I mainly viewed GitHub as a website for professional programmers. I now understand that it can also be used to store and share notebooks, code, documentation, and complete workflows. I still have much more to learn about version control and GitHub's full capabilities, but I can see it becoming useful for saving reusable scripts and showing how a data process was completed. In a professional setting, I would need to be careful not to upload confidential company or project information, but I could use it for personal practice projects, nonconfidential templates, and examples of my technical skills.

If the course had more time, I would have liked to learn more about connecting Python with Excel, automatically processing multiple files, creating dashboards, and producing more advanced visualizations. I would also be interested in learning how Python could compare two versions of a dataset and clearly identify what changed. That type of workflow could be useful in estimating when comparing drawing sets, bid information, or updated project data.

I plan to continue learning Python, although I expect to use it alongside Excel rather than as a complete replacement. My next step would be to practice with small, realistic tasks instead of trying to build a large program immediately. I could continue using DataCamp, review the notebooks from this course, and experiment with nonconfidential sample datasets related to estimating and business analysis. Overall, this course gave me a better understanding of how programming can make data cleaning more consistent, repeatable, and transparent. It also made Python feel more approachable and showed me how it could become a useful addition to my business and analytical skill set.